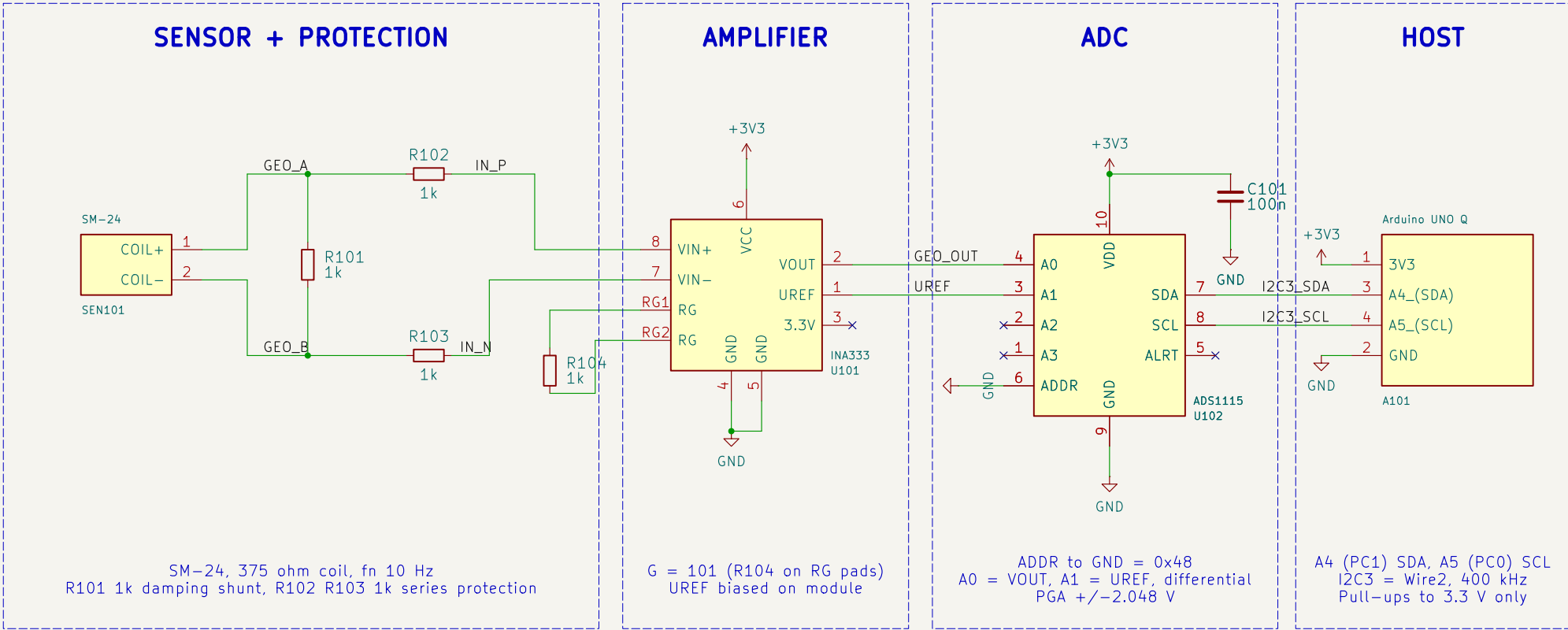


EleTect X – GEOPHONE ANALOG FRONT-END BOARD

SM-24 geophone > INA333 instrumentation amplifier > ADS1115 16-bit ADC > Arduino UNO Q (I2C3)



Sheet 1 of 4 | designators 1xx
Drawn by Abhinav Krishna – EleTect X
3V3 supply. R101–R104 1k, C101 100n
SM-24 > INA333 (G=101) > ADS1115 (0x48) > UNO Q
EleTect X | Designed by Abhinav Krishna

Sheet: /Geophone/
File: geophone.kicad_sch

Title: Geophone Analog Front-End Board

Size: A4 Date: 2026-10-04

KiCad E.D.A. 10.0.5

Rev: A

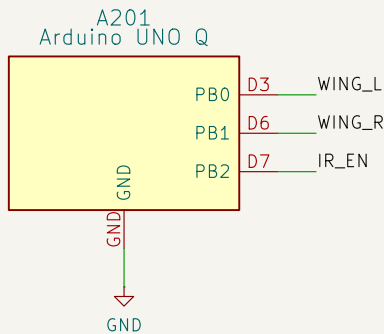
Id: 1/5

EleTect X – LED WING + IR ILLUMINATOR MOSFET DRIVERS

Arduino UNO Q GPIO > gate resistor > IRLZ44N low-side switch > LED wings and 940 nm IR board

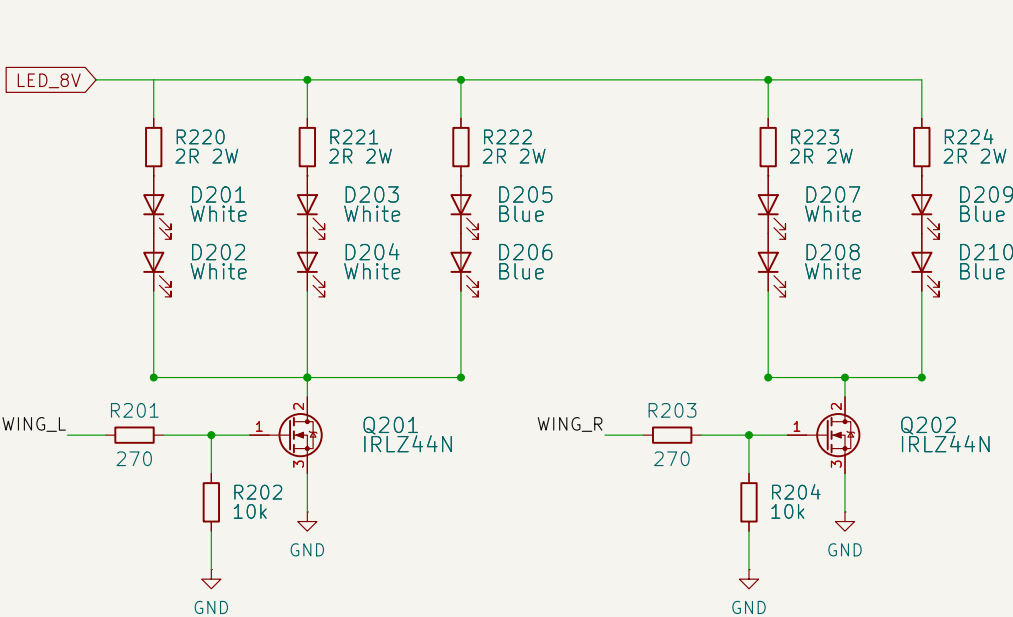
LED_8V is the 8 V LED buck output and IR_VBAT the fused battery bus, both from the power sheet; battery, fuses and switch live there

HOST (Arduino UNO Q)



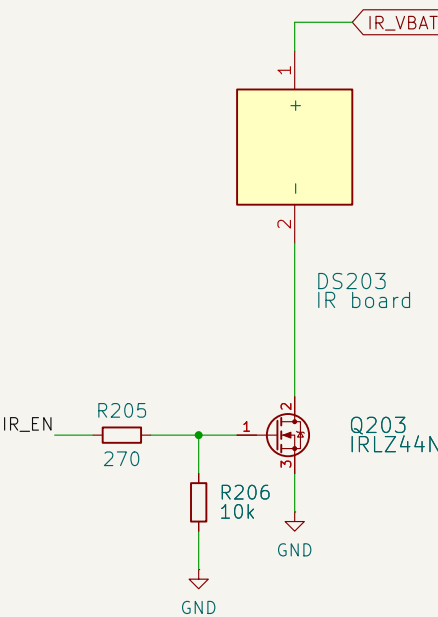
GPIO outputs, 3.3 V logic
D3/PB0 is not 5 V tolerant:
no pull-up above 3.3 V on it

LED WINGS (white + blue)



Fed from the 8 V LED buck on the power sheet; two LEDs in series per string (6.6 V)
2R 2 W sets about 700 mA per string at Vf 3.3 V; trim on the bench
270R limits gate inrush, 10k pulldown holds the MOSFET off at boot

IR ILLUMINATOR (940 nm)



Sheet 2 of 4 | designators 2xx
Drawn by Abhinav Krishna – EleTect X
Rails LED_8V (8 V buck) and IR_VBAT (fused battery bus) come from the power sheet
UNO Q GPIO > 270R > IRLZ44N low-side switch > load; 10k gate pulldown

EleTect X | Designed by Abhinav Krishna

Sheet: /LED and IR/
File: led_ir.kicad_sch

Title: LED Wing + IR Illuminator MOSFET Drivers

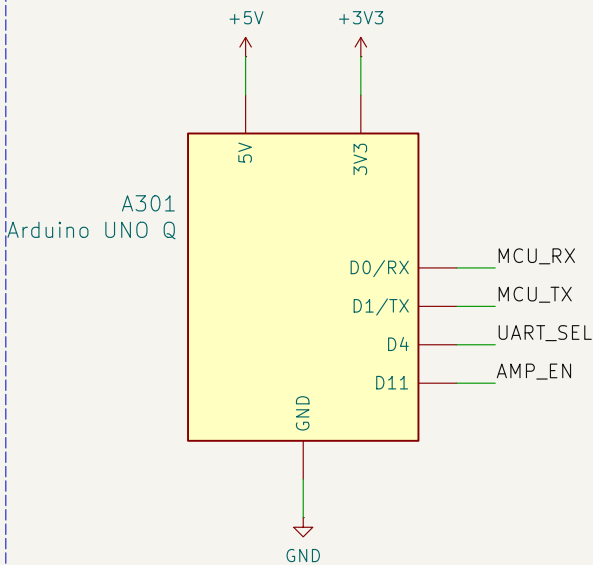
Size: User Date: 2026-10-04
KiCad E.D.A. 10.0.5

Rev: A
Id: 2/5

EleTect X – DFPLAYER PRO + HORN AMP + SHARED UART

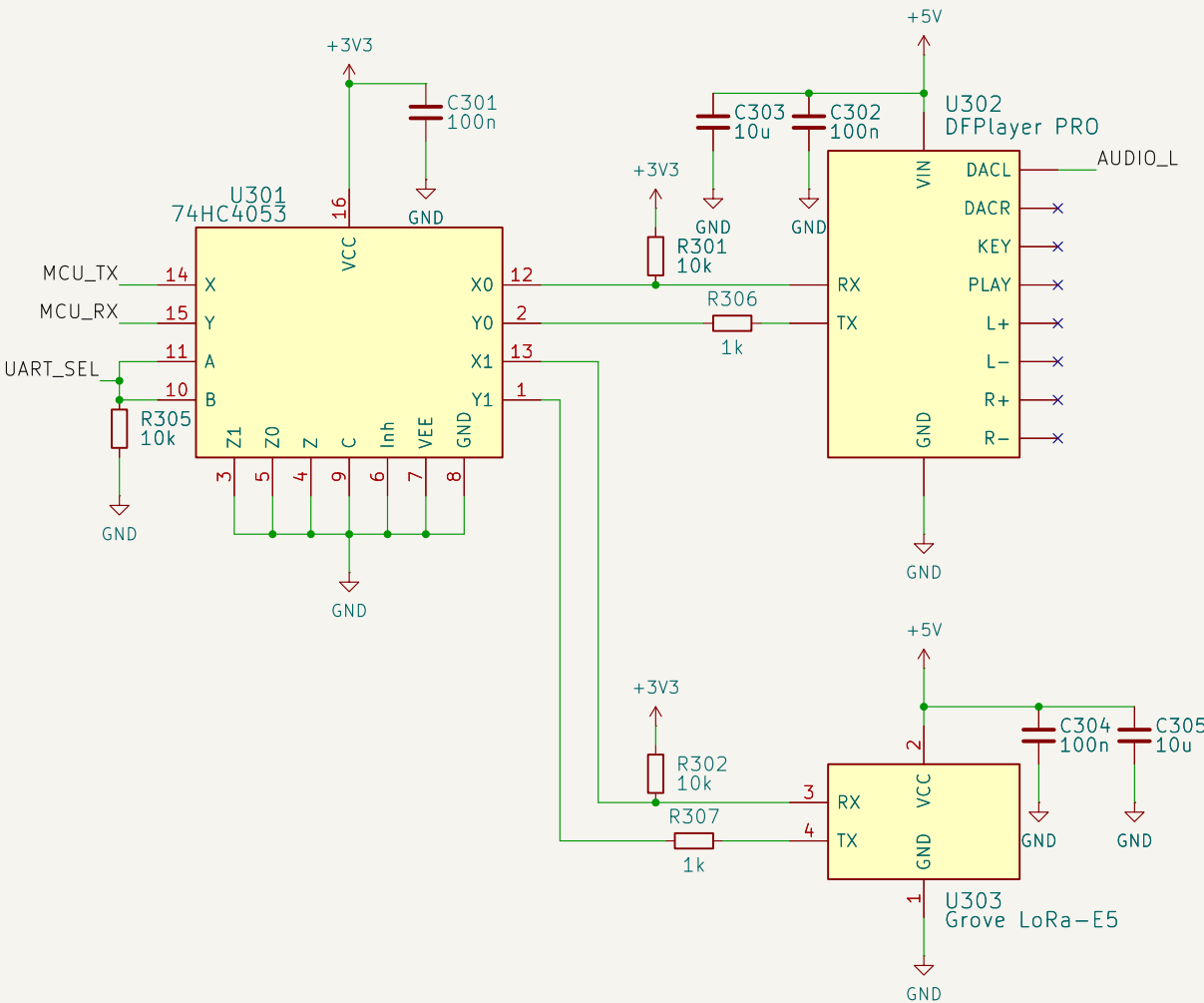
UNO Q USART1 > 74HC4053 > DFPlayer PRO / LoRa-E5; DFPlayer DACL > XH-M543 amp > Ahuja SUH-15 horn
Horn amp runs from the raw 12.8 V battery bus; DFPlayer PRO and LoRa-E5 run from the UNO Q 5 V pin

HOST (UNO Q)



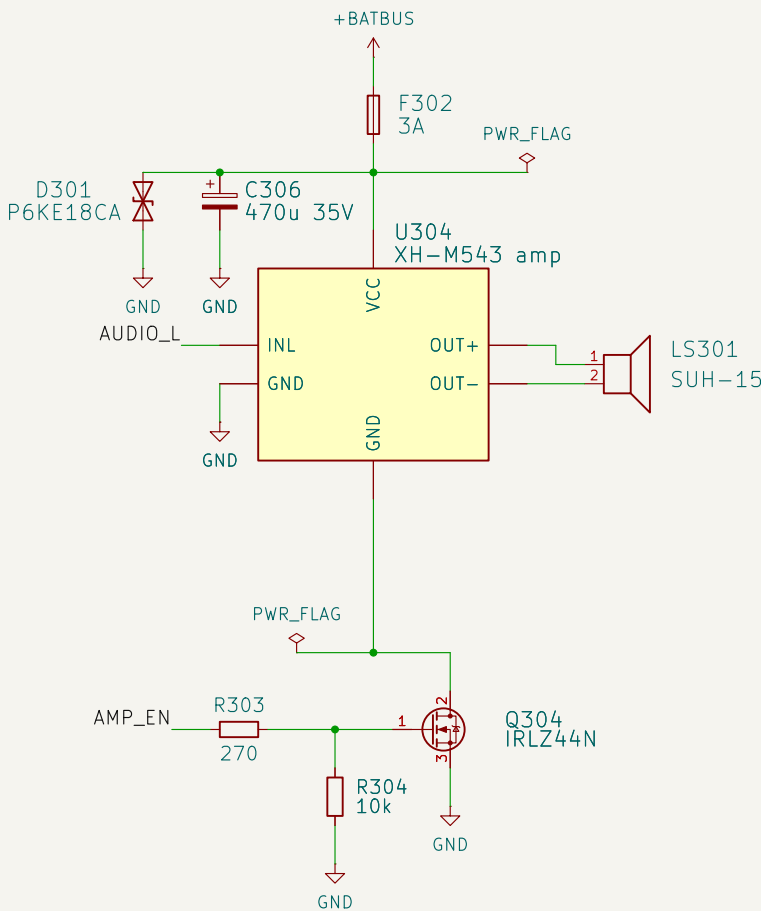
D0 / D1 = USART1 (3.3 V pins)
D4 select: LOW = DFPlayer PRO, HIGH = LoRa-E5
D11 gate HIGH = amp powered

SHARED UART (74HC4053 + DFPlayer PRO + LoRa-E5)



74HC4053 runs at 3.3 V; the horn always wins the shared port (ADR 0029)
DFPlayer PRO 115200 baud, LoRa-E5 9600 baud: each switch re-opens Serial1
10k pull-ups hold the deselected RX line idle high
R305 pulls D4 low at reset: DFPlayer PRO selected by default
1k series R306 / R307 limit clamp current if a module TX swings above 3.3 V
Unused 4053 pins (C, Inh, VEE, Z, Z0, Z1) are tied to GND

HORN (XH-M543 amp + Ahuja SUH-15)

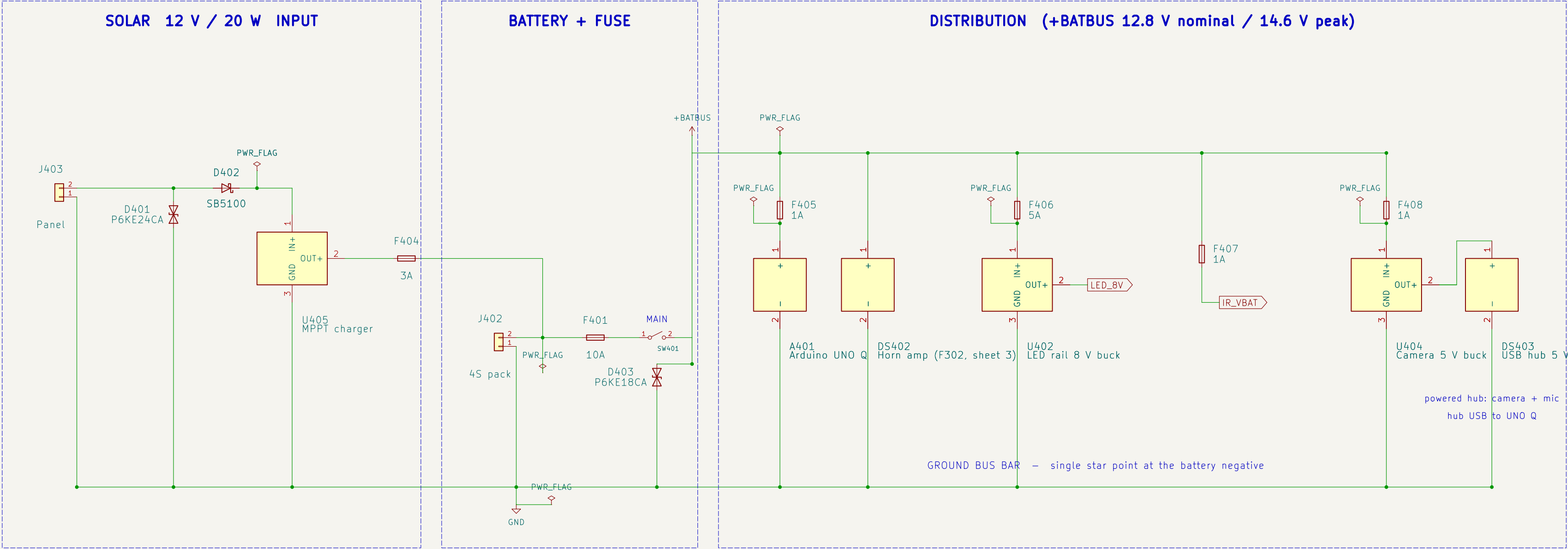


Q304 switches the amp ground: gate HIGH = amp on
F302 3A fuse, D301 TVS and C306 bulk cap protect the amp supply
Turn VOL_L fully down before first power-on
DACL feeds INL, not the DFPlayer L+ / L- pins
Amp joins its input and power grounds on the board:
check it goes quiet with D11 LOW (DFPlayer GND is a 2nd return)

EleTect X – POWER: SOLAR, BATTERY, BUCKS

20 W panel > P6KE24CA TVS > SB5100 > MPPT controller (LiFePO4 mode) > 4S LiFePO4 > 10 A fuse > switch > +BATBUS > branches

One bus bar at the battery negative carries every return; the LED rail and the camera hub each have their own buck



Sheet 4 of 4 designators 4xx		
Drawn by Abhinav Krishna – EleTect X		
Single star ground at the battery negative; LED rail and camera hub on separate bucks		
Solar > MPPT controller (LiFePO4) > 4S LiFePO4 > 10 A fuse > switch		
EleTect X Designed by Abhinav Krishna		
Sheet: /Power/		
File: power.kicad_sch		
Title: Power: Solar, Battery, Bucks		
Size: User	Date: 2026-10-04	Rev: A
KiCad E.D.A. 10.0.5	Id: 4/5	